

1 Notation & Perturbation Method

Big $\mathcal{O}$: $f(z) = \mathcal{O}(g(z))$ as $z \rightarrow z_0$ if $\lim_{z \rightarrow z_0} \frac{f(z)}{g(z)} \text{ bdd} \Leftrightarrow |f(z)| \leq A|g(z)|$ as $z \rightarrow z_0$

Small o : $f(z) = o(g(z))$ as $z \rightarrow z_0$ if $\lim_{z \rightarrow z_0} \frac{f(z)}{g(z)} = 0 \Leftrightarrow f(z) \ll g(z)$ as $z \rightarrow z_0$

Asymptotic $\sim$: $f(z) \sim g(z)$ as $z \rightarrow z_0$ if $\lim_{z \rightarrow z_0} \frac{f(z)}{g(z)} = 1$ **ps:** $e^{-x} = o(x^{-n}), \forall n > 0$ as $x \rightarrow \infty$

Asymptotic Sequence: Seq $\{\varphi_n(z)\}$ is asy. if $\forall n \in \mathbb{N} \varphi_{n+1}(z) = o(\varphi_n(z))$ as $z \rightarrow z_0$

Asymptotic Expansion: $f(z) \sim \sum_{n=0}^M a_n \varphi_n(z)$ as $z \rightarrow z_0$ to M if $\phi_n(z)$ is asy. seq. and

$f(z) - \sum_{n=0}^N a_n \varphi_n(z) = o(\varphi_N(z)) = \mathcal{O}(\varphi_{N+1}(z)), \forall N \leq M$ as $z \rightarrow z_0$

$f(x, \varepsilon) = \sum_{n=0}^N a_n(x) \varepsilon^n$ uniformly in $x \in I$ as $\varepsilon \rightarrow 0$ if ε 与 x 无关 (i.e. sup $|f(x, \varepsilon) - \sum_{n=0}^N a_n(x) \varepsilon^n| = o(\varepsilon^N)$)

Singular|Regular - Perturbation Method: a. Let $\varepsilon = 0$ solve $f(x, 0) = 0$.

b.i. 如果 f 结构变了 (阶数降低), Let $x = \varepsilon^{-\alpha} X$, PDB 找 α . c. 假设 $X = a_0 + a_1 \varepsilon + a_2 \varepsilon^2 + \dots$ 代入求解
b.ii. 如果 f 结构不变/变, 但有重根 x_0 , Let $x = x_0 + \varepsilon^\alpha X$, PDB 找 α . $\uparrow \beta$ 看化简后的 X 方程得出

Principle of dominant balance (PDB): For Singular Perturbation $p(x, \varepsilon) = 0$, Let $x = \varepsilon^\alpha X$, 代入选择 α :
对于 ε 来说 a. 至少有两项阶数相同且最低; b. 其他项阶数均更高. (即可忽略)

2 Integral By Parts, Laplace's Method & Method of Steepest Descents

Taylor: $f(x) = \sum_{n=0}^N \frac{f^{(n)}(x_0)}{n!} (x - x_0)^n + \frac{f^{(N)}(\xi)}{N!} (x - x_0)^N \Rightarrow f(x) \sim \sum_{n=0}^{\infty} \frac{f^{(n)}(x_0)}{n!} (x - x_0)^n$ as $x \rightarrow x_0$

Exp. Integral: $E_1(x) = \int_x^\infty \frac{e^{-t}}{t} dt$, $(x \rightarrow 0 \text{ 发散})$ $E_1(x) \sim e^{-x} \sum_{n=0}^{\infty} \frac{(-1)^n}{x^{n+1}}$, $x \rightarrow \infty$

Gamma Func: $\Gamma(x) = \int_0^\infty t^{x-1} e^{-t} dt$ $\Gamma(x+1) = x\Gamma(x)$ and $\Gamma(n) = (n-1)!$ $\Gamma(\frac{1}{2}) = \sqrt{\pi}$

Incomplete Gamma Func: Lower: $\gamma(a, x) = \int_0^x t^{a-1} e^{-t} dt$ Upper: $\Gamma(a, x) = \int_x^\infty t^{a-1} e^{-t} dt$
 $\gamma(a, x) = \Gamma(a) - \Gamma(a, x)$ $\Gamma(a, x) \sim e^{-x} x^{a-1} \sum_{n=0}^{\infty} \frac{\Gamma(a)}{\Gamma(a-n)} x^{-n}$, $x \rightarrow \infty$

Laplace's Method: For $I(x) = \int_a^b e^{x f(t)} g(t) dt$ as $x \rightarrow \infty$ $e^{x[f(t_0) + \frac{f''(t_0)}{2}(t-t_0)^2 + \dots]}$ $g(t_0) dt$

1. If $f(t)$ has unique max at $t = t_0 \in (a, b) \Rightarrow I \sim \int_{-\infty}^{\infty} e^{x f(t_0) + \frac{f''(t_0)}{2}(t-t_0)^2 + \dots} g(t_0) dt$
(留 n 项非常数展开为 n term approximation) (ps: max $\leftrightarrow f'(t_0) = 0, f''(t_0) < 0$)

Useful Tool: Now: $I \sim e^{x f(t_0)} \int_{-\infty}^{\infty} e^{x[\frac{f''(t_0)}{2}(t-t_0)^2 + \dots]} g(t_0) dt \Rightarrow \text{Let } \tau^2 = -[\frac{1}{2} f''(t_0)(t-t_0)^2 + \dots]$
Assume $t = \tau + b\tau^2 + c\tau^3 + \dots$ as $\tau \rightarrow 0 \Rightarrow$ 代入换元式子求系数 $\Rightarrow dt = (1 + 2b\tau + 3c\tau^2 + \dots) d\tau$

2. **Exact Substitution:** 由 1 的基础, 得到 $f(t)$ 在 t_0 处的展开式, 换元令 $f(t)$ 就等于展开式前几项求解.

3. If $g = g(t, x): e^{x f(t)} g(t, x) = e^{x f(t) + \ln(g(t, x))}$, 看 $x f(t) + \ln(g(t, x))$ 的最大值, 类似 1 处理.
- If $t_0 = f(x)$, 换元令 $t = f(x)$.

Method of Steepest Descents: For $I(x) = \int_C e^{x f(z)} g(z) dz$ as $x \rightarrow \infty$. f 容易避开 singular point

If z_0 is saddle point: $f'(z_0) = 0, f''(z_0) \neq 0$ $f(z)$ Taylor at $z_0: f(z) = f(z_0) + \frac{f''(z_0)}{2}(z-z_0)^2 + \dots$

1. Let $z = z_0 + e^{i\theta} y$ (θ 选让 $f''(z_0)e^{2i\theta}$ 为负实数) $\rightarrow I(x) \sim \int_{-\infty}^{\infty} e^{x[f(z_0) - \frac{|f''(z_0)|}{2}y^2 + \dots]} g(z_0)e^{i\theta} dy$

2. If f 不容易避开 singular point: a. $\text{Im}(f(z)) = \text{Im}(f(z_0))$ and $z_0 \in C_0$ to find steepest descent path C_0

3. b. 画图看哪些 singular points 在 $C \rightarrow C_0$ 变形路径上, 计算 residues. c. $I = 1$ 部分 + $2\pi i \sum$ residues.
4. **More than one saddle point:** i. 如果通过积分区域可以舍弃, 则舍弃 ii. saddle points: z_k , 取能让 $\text{Re}(f(z_k))$ 最小的.
iii. 如果多个 saddle points 都满足 $\text{Re}(f(z_k))$ 最小, 则都要计算后叠加.

5. **Exact Substitution:** 由 1 的基础, 得到 $f(z)$ 在 z_0 处的展开式, 换元令 $f(z)$ 就等于展开式前几项求解.

6. **Remark:** 类似 Laplace 3 一样, 可以对 $x f(x)$ 一个整体处理, 必要时 $z_0 = f(x) \rightarrow$ 换元 $z = f(x)$ s.

Residue: If z_0 pole of order m of f , $\Rightarrow \text{Res}(f, z_0) = \frac{1}{(m-1)!} \lim_{z \rightarrow z_0} \frac{d^{m-1}}{dz^{m-1}} [(z-z_0)^m f(z)]$

Method of Stationary Phase: $\cos(x f(\theta)) \rightarrow \text{Re}(e^{i x f(\theta)})$ 回到 Steepest Descents, 同理 $\sin(x f(\theta))$

3 Perturbation Methods in ODEs

Regular Perturbation: For $L(y, \varepsilon) = 0$, let $y(x, \varepsilon) = y_0(x) + \varepsilon y_1(x) + \varepsilon^2 y_2(x) + \dots$
 $\Rightarrow$ a. 代入 $\mathcal{O}(\varepsilon^n) \rightarrow$ b. Boundary/Initial Condition 求解 $y_n(x) \rightarrow$ a.

Singular Perturbation: e.g. $\varepsilon y'' + p(x)y' + q(x)y = 0, y(a) = A, y(b) = B, a < b$, 大致步骤:

- Assume boundary layer at $x = a|b$. Let $x = a + \delta(\varepsilon)X$ or $x = b - \delta(\varepsilon)X$
- Assume inner $Y(X) = y(x)$, chain rule 求导代入原 ODE $\rightarrow$ Domain balance 找出 $\delta(\varepsilon) = \varepsilon^\alpha$
Recall PDB: a. 至少有两项阶数相同且最低; b. 其他项阶数均更高.
- Assume $Y = Y_0(X) + \varepsilon^\beta Y_1(X) + \varepsilon^{2\beta} Y_2(X) + \dots$ 代入求解 $\rightarrow$ 边界/初始条件求解 $Y_0(X)$
- Matching:** Consider $X \rightarrow \infty | x \rightarrow a|b$. If limits not exist (除非 X) $\rightarrow$ boundary layer 在另一个端点 $\rightarrow 1$
If limits exist, boundary layer 正确 $\rightarrow$ 渐进 outer (X), 算 inner 极限 (X); 算不了保留 $X \rightarrow x$ 和 X 的关系代入 $\rightarrow$ 比较系数.

Remark 1: Outer solution 只能用在 away from boundary layer 的区域 (包括边界点).

Remark 2: Inner solution 只能用在 boundary layer 附近的区域 (包括边界点).

Useful Lemma: For $\varepsilon y'' + p(x)y' + q(x)y = 0, y(a) = A, y(b) = B, a < b, p(x) \neq 0$ in $[a, b]$ then
Has boundary layer at $x = a$ if $p > 0$; Has boundary layer at $x = b$ if $p < 0$.

4 WKB Theory & Solving PDE

WKB: For $u'' + k^2 q(x)u = 0, k \gg 1$ (or $k^2 q \gg 1$) we have: $u \sim A(x)e^{ik\phi(x)}$. Substitution can get:

If $q(x) < 0: u(x) \sim A_+ |q|^{-\frac{1}{4}} e^{-\frac{k}{2} \int_{x_0}^x \sqrt{|q(t)|} dt} + A_- |q|^{-\frac{1}{4}} e^{-\frac{k}{2} \int_{x_0}^x \sqrt{|q(t)|} dt}$ (Exp. Decay/Growth)

If $q(x) > 0: u(x) \sim A_+ q^{-\frac{1}{4}} e^{i \frac{k}{2} \int_{x_0}^x \sqrt{q(t)} dt} + A_- q^{-\frac{1}{4}} e^{-i \frac{k}{2} \int_{x_0}^x \sqrt{q(t)} dt}$ (Oscillation)

Turning Point: If $q(x_0) = 0$, then x_0 is a WKB turning point.

Method of Characteristics|IC: For $u_t + c(x, t)u_x = r(x, t, u)$ with IC $u(x, t_0) = f(x)$:

- In characteristic form: $\frac{du}{dt} = r(x, t, u)$ on $\frac{dx}{dt} = c(x, t) \rightarrow$ 解 ODEs $\rightarrow u, x$ 各产生一个未知常数 A, B .
- Assume characteristic curve (solution of $\frac{dx}{dt} = c(x, t)$) $x(t = t_0) = \xi$ SO $u(t = t_0) = f(\xi)$:
 $\rightarrow$ get $u(t = t_0) = f(\xi)$ on $x(t = t_0) = \xi \rightarrow$ solve A, B .
- Substitute A, B back to get u .

Method of Characteristics|BC: For $u_t + c(x, t)u_x = r(x, t, u)$ with BC $u(x_0, t) = f(t)$:

- In characteristic form: $\frac{du}{dt} = r(x, t, u)$ on $\frac{dx}{dt} = c(x, t) \rightarrow$ 解 ODEs $\rightarrow u, t$ 各产生一个未知常数 A, B .
- Assume characteristic curve (solution of $\frac{dx}{dt} = c(x, t)$) $t(x = x_0) = \xi$ SO $u(x = x_0) = f(\xi)$:
 $\rightarrow$ get $u(x = x_0) = f(\xi)$ on $t(x = x_0) = \xi \rightarrow$ solve A, B .
- Substitute A, B back to get u .

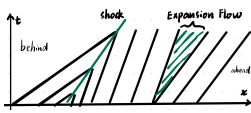
Special Cases of Method of Characteristics:

- 如果 Characteristic curve 和初始曲线/边界条件相交 $\rightarrow$ ill-posed & if not unique $\rightarrow$ no solution.
- For $u_t + c(u)u_x = 0$:** If a initial jump $u(x, t_0) = u_L$ for $x < b$ and $u(x, t_0) = u_R$ for $x > b$
If $c(u_L) < c(u_R) \rightarrow$ expansion wave; If $c(u_L) > c(u_R) \rightarrow$ shock wave.
- For $u_t + c(u)u_x = 0, u(x, t_0) = f(x)$:** The breaking time $t_b = t_0 + \min_\xi \frac{-1}{F'(\xi)}$ where $F(\xi) = c(f(\xi))$
Break time is $\frac{du}{dx} = \infty$ (i.e. 分母为 0) 时的最早时间. (通过 chain rule 得到) || If $\min_\xi F'(\xi) < 0 \rightarrow$ shock

Expansion: For $u_t + c(u)u_x = 0, u(x, t_0) = u_L | u_R$
1. 正常解 $\rightarrow \xi$ 边界值 $\xi_0 \rightarrow$ 特征曲线 $x(t; \xi)$ 在 ξ_0 边界情况:
 $x = c(u_L)(t - t_0) + \xi_0$ 和 $x = c(u_R)(t - t_0) + \xi_0$

2. $\Rightarrow \frac{x - \xi_0}{t - t_0} = c(u)$ for $c(u_L)(t - t_0) + \xi_0 < x < c(u_R)(t - t_0) + \xi_0$
 $\Rightarrow u = c^{-1}[\frac{x - \xi_0}{t - t_0}]$

Conservation Law: $u_t + c(u)u_x = 0: u_t + [f(c(u))u]_x = 0$



Shock Wave $|u_L, u_R$: For $u_t + c(u)u_x = 0, u(x, t_0) = u_L$ for $x < b$ and $u(x, t_0) = u_R$ for $x > b$.

- 正常解 $\rightarrow$ In conservation form: $u_t + [f(c(u))u]_x = 0$
shock velocity $\frac{ds}{dt} = \dot{s} = \frac{[q]}{[\rho]}$ where $q = \int c(u)du$ and $\rho = u$; $[u] := u_{\text{ahead}} - u_{\text{behind}}$.
其中: $u_{\text{ahead}} = u_R, u_{\text{behind}} = u_L$
- Solve $\dot{s}$. As shock starts at $x = b$ at $t = t_0, s(t = t_0) = b$ get unknown constant.
- Then solution $u = u_L$ for $x < s(t)$ and $u = u_R$ for $x > s(t)$.

Shock Wave $|f(x)$: For $u_t + c(u)u_x = 0, u(x, t_0) = f(x)$ with $t_b = t_0 + \min_\xi \frac{-1}{F'(\xi)}$ where $F(\xi) = c(f(\xi))$.

- 正常解 $\rightarrow$ In conservation form: $u_t + [f(c(u))u]_x = 0$
shock velocity $\frac{ds}{dt} = \dot{s} = \frac{[q]}{[\rho]}$ where $q = \int c(u)du$ and $\rho = u$; $[u] := u_{\text{ahead}} - u_{\text{behind}}$.
其中: $u_{\text{ahead}} = \lim_{x \rightarrow s(t)^+} u(x, t), u_{\text{behind}} = \lim_{x \rightarrow s(t)^-} u(x, t)$ (which is calculated by characteristics method)
- Solve $\dot{s}$. As shock starts at $x(t = t_b) = t_b, s(t = t_b) = x(t = t_b)$ get unknown constant.
ps: i.e. $x(t = t_b) = \xi_+ + F(\xi_+)(t_b - t_0)$, 其中 ξ_+ 是使得 $t_b = t_0 + \min_\xi \frac{-1}{F'(\xi)}$ 的 ξ .

Remark: 可能 Expansion and Shock 同时存在且衔接一起. (看 u 解出来后 x 范围 $\rightarrow$ 某些范围是否可被压缩)
此时 Solve $\dot{s}$ 后, 用 $t_b =$ 压缩点时间, $x =$ 原先 shock 时 x 的特征线代入 t_b 的解 $\rightarrow$ 求未知常数.

5 Appendix

x, x^2, x^3 as $x \rightarrow 0, \quad x^{-1}, x^{-2}, x^{-3}$ as $x \rightarrow \inf$

5.1 ODE

$y' + P y = Q: \mu y = \int Q(x) \mu dx + C, \quad \mu(x) = e^{\int P(x) dx}$

$y'' + k^2 y = 0: y = A \cos(kx) + B \sin(kx) \quad y'' - k^2 y = 0: y = A e^{kx} + B e^{-kx}, k > 0$

5.2 Integral

Gaussian Integral: $\int_{-\infty}^{\infty} e^{-ax^2} dx = \frac{\sqrt{\pi}}{\sqrt{a}} \quad \int_0^\infty e^{-x t^n} t^m = \frac{1}{n} \Gamma(\frac{m+1}{n}) x^{-\frac{m+1}{n}}$

Integral: $\int \tan x = -\ln |\cos x| = \ln |\sec x| \quad \int \cot x = \ln |\sin x| = -\ln |\csc x|$

Techniques: a. e.g. $\int e^{at} \cos(bt) dt = \text{Re}(\int e^{(a+ib)t} dt)$

5.3 Expansion

Binomial Expansion: $(1+x)^\alpha = 1 + \alpha x + \frac{\alpha(\alpha-1)}{2!} x^2 + \dots$ as $x \rightarrow 0$

For $x \rightarrow a$: Taylor at $x = a: f(x) \sim f(a) + f'(a)(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \dots$ as $x \rightarrow a$

Taylor $|x \rightarrow 0$: $f(x+h) \sim f(x) + f'(x)h + \frac{f''(x)}{2!}h^2 + \dots$ as $h \rightarrow 0$

$\sin(x) \sim x - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots$ as $x \rightarrow 0 \quad e^x \sim 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots$ as $x \rightarrow 0$

$\cos(x) \sim 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \dots$ as $x \rightarrow 0 \quad \ln(1+x) \sim x - \frac{x^2}{2} + \frac{x^3}{3} - \dots$ as $x \rightarrow 0$

$\sinh(x) \sim x + \frac{x^3}{3!} + \frac{x^5}{5!} + \dots$ as $x \rightarrow 0 \quad \cosh(x) \sim 1 + \frac{x^2}{2!} + \frac{x^4}{4!} + \dots$ as $x \rightarrow 0$

$\tan(x) \sim x + \frac{x^3}{3} + \frac{2x^5}{15} + \dots$ as $x \rightarrow 0 \quad \cot(x) \sim \frac{1}{x} - \frac{x}{3} + \frac{x^3}{45} - \dots$ as $x \rightarrow 0$

For $x \rightarrow \infty$: $\arctan(x) \sim \frac{\pi}{2} - \frac{1}{x} + \frac{1}{3x^3} - \dots$ as $x \rightarrow \infty$

$\ln(a+x) = \ln x + \ln(1 + \frac{a}{x}) \sim \ln x + \frac{a}{x} - \frac{a^2}{2x^2} + \dots$ as $x \rightarrow \infty$

$\frac{1}{a+x} = \frac{1}{x} \cdot \frac{1}{1+\frac{a}{x}} \sim \frac{1}{x} - \frac{a}{x^2} + \frac{a^2}{x^3} - \dots$ as $x \rightarrow \infty$

sin|cos: $a \sin(x) + b \cos(x) = \sqrt{a^2 + b^2} \sin(x + \varphi) \quad a \cos(x) + b \sin(x) = \sqrt{a^2 + b^2} \cos(x - \varphi), \varphi$ 过 (a, b) .

$-\sin(x) + \sin(y) = 2 \sin(\frac{x+y}{2}) \cos(\frac{x-y}{2}) | \sin(x) + \cos(y) = \frac{1}{2} [\sin(x+y) + \sin(x-y)] \quad S+S=SC, S-S=CS, C+C=CC, S-S=SS$

Inequalities: $\ln(x+1) < x, \quad e^x > x+1, \quad (1+x)^a < e^{ax}$

5.4 Other

Nabla: $\nabla^2 u = u_{xx} + u_{yy} + u_{zz} + \dots \quad \nabla \cdot \mathbf{F} = \frac{\partial F_x}{\partial x} + \frac{\partial F_y}{\partial y} + \frac{\partial F_z}{\partial z}$

Binomial Expansion: $(a+b)^n = \sum_{k=0}^n \binom{n}{k} a^{n-k} b^k \quad \binom{n}{k} = \frac{n!}{k!(n-k)!} \quad (1+x)^n = \sum_{k=0}^n \binom{n}{k} x^k$